

Data Science Intern Assignment

Problem Definition:

For this assignment I chose a dataset labeled “Cyberbullying”, which consists of 11000 user generated text comments, categorized binary as either cyberbullying (1) or non-cyberbullying (0). The dataset has a balanced ratio (1:1) between the categories, making it well-suited for classification tasks. This would allow me to tackle relevant challenges regarding detection of cyber-bullying in user-generated text comments. *The goal is to build a machine learning model for detecting cyberbullying in user-generated text.*

The goal for this assignment is to develop a proof-of-concept model that can effectively classify text as either cyberbullying or non-cyberbullying. This involves applying Natural Language Processing (NLP) techniques to preprocess and analyze the text data, followed by training multiple classification models to assess their performance, evaluating the best model.

Findings & Key insights:

During EDA, when looking at the histogram, I was assuming that cyberbullying texts are shorter in length compared to non-cyberbullying texts since the spread for non-cyberbullying was wider. Looking at the mean and median length (also t-test and p-value), I was pleasantly surprised that the results were counterintuitive and therefore had maybe something to do with the outliers’ impact on the dataset. This showed that although visual representations in ML can be beneficial to conceptualize data, it does not reflect the underlying behavior of the dataset. One way to address it is by handling outliers as noise. However, removing outliers can create biases and may not be representative of real-world cyberbullying datasets, which are often diverse and unclean.

When modeling, I used logistic regression (also w/hyperparametric tuning) and random forest. Here are the simplified results:

Values	Logistic Regression	Logistic Regression (w/parametric tuning)	Random Forest
Accuracy (%)	72.16%	72.39%	72.03%
Precision (0)	0.71	0.71	0.69
Recall (0)	0.77	0.77	0.81
F1-score(0)	0.74	0.74	0.75
Precision (1)	0.74	0.74	0.75

Recall (1)	0.67	0.67	0.63
F1-score(1)	0.70	0.70	0.69
Precision (macro avg)	0.72	0.72	0.73
Recall (macro avg)	0.72	0.72	0.72
F1-score (macro avg)	0.72	0.72	0.72
Confusion Matrix	[[874 254] [364 728]]	[[874 254] [364 728]]	[[914 214] [407 685]]

Applying hyperparametric tuning slightly improved the accuracy of the logistic regression model, being the overall best model during evaluation. But the change is significantly small, which suggests that tuning the C and solver had little impact on the model performance. The unchanged precision, recall, F1-score, and confusion matrix indicate that hyperparametric tuning did not improve the model. Since logistic regression is considered a stacking model, stacking it with other models could improve the overall performance of the model. Due to time constraints, I was not able to explore stacking.

The Random Forest model has insignificantly less accuracy than both of the logistic models, indicating that none of the models provide a significant advantage in classification performance. The higher recall for Non-cyberbullying indicates that Random Forest is better at identifying non-cyberbullying texts, which we can see at the reduced False positives on its confusion matrix. However, it misses more cyberbullying comments compared (see false-negatives) to the other models. An improvement for the model could have been to use word embeddings for feature extraction instead of TF-IDF, since word embeddings capture the semantic meaning of words rather than just considering the word frequency of the document.

When testing the final model, it does seem to distinguish words between cyberbullying and non-cyberbullying. However, it does misclassify a number of non-cyberbullying comments as cyberbullying (false-positives) and also fails to detect some actual cyberbullying instances. This aligns with the confusion matrix, suggesting room for improvement in precision and recall. As mentioned earlier, adding more techniques such as word embeddings or stacking could improve the overall performance, which I wish to explore in the future.